Electrical machine, mechanical power equation of electrical machine, hysteresis and eddy current loss, Linear DC machine, starting problem, Linear DC machine as motor and Generator
Formulae only: Faraday's law, Induced Force on a Wire, Induced Voltage on a Conductor and all of the related maths

Importance of motor sizing, how to choose the right motor, graphs on motor starting, motor sizing for variable load based on service factor and related math problem, motor duty classes with figures, motor derating, energy efficient motor.

Transformer and it's role in power system, transformer working principle, transformer equivalent circuit, equivalent circuit referred to primary, exact equivalent circuit, approximate equivalent circuit, eddy current loss, hysteresis loss, what are the causes for transformer being rated in KVA, conditions for maximum efficiency, No load test, short circuit test.
Math: 32.1 - 32.4, 32.21 (less important)
32.15, 32.16, 32.27, Spring - 25 math (***most Important)

DC Generator and dc motor working principle, action of commutator/commutation, flemming's right and left hand rule, Math 26.3, 29.1